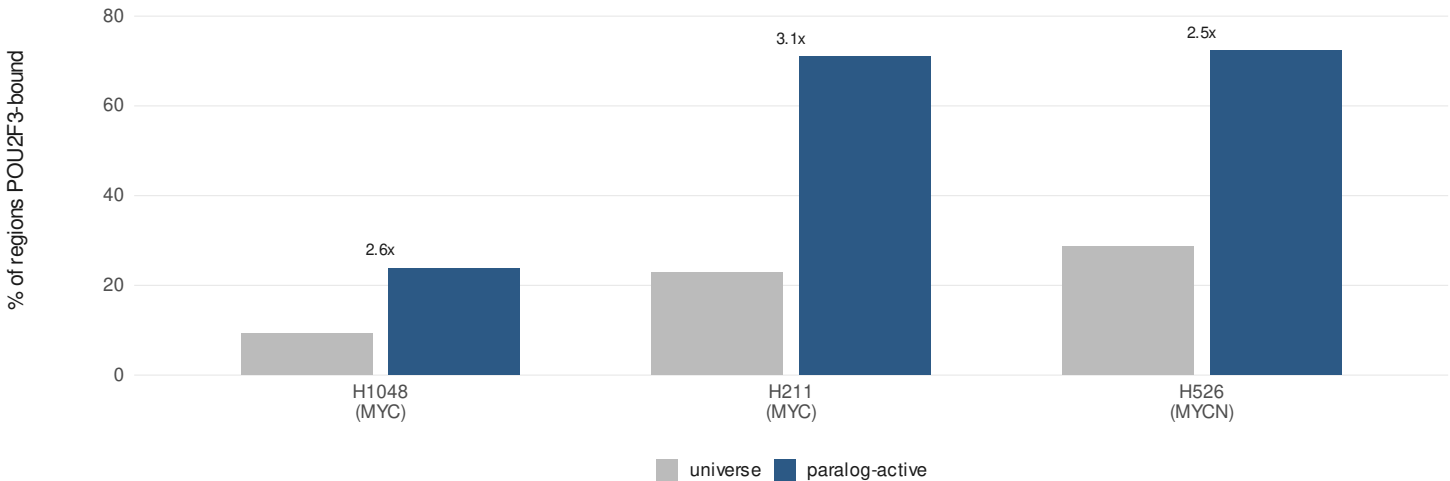


The lineage confound is present at chromatin, cell-line and gene level — not only in tumour scores

The negative result in Figure 1 does not depend on how regulon scores were computed. Neuroendocrine lineage state is entangled with MYC-family regulatory occupancy in the cell lines the regulons were built from, and survives every level at which it was tested.

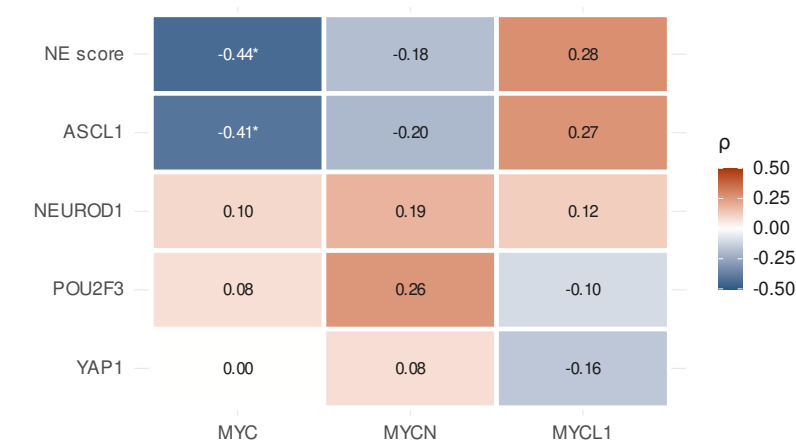
A · The confound is already in the chromatin

POU2F3-bound fraction: paralog-active regions vs the universe
labels give the enrichment; all three $p < 1e-15$



B · MYC tracks lineage in cell lines too

Spearman rho, CCLE SCLC lines (n=59); * is FDR < 0.05



C · Survivors are not enriched in the regulons

genes still associated with the paralog after lineage adjustment
grey text: enrichment of survivors inside vs outside the regulon

